(c) Explain the observations seen in experiment 3. Use these to name element X. [3]

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(d) In experiment 4 the acid reacts with all the hydroxide and carbonate ions.

$$H^+ + OH^- \longrightarrow H_2O$$

$$2H^+ + CO_3^{2-} \longrightarrow H_2O + CO_2$$

Calculate the value of **d**, the number of hydroxide ions present in each formula unit. [4]

**d** = ..............................

Examiner only

(e) The compound has 56 atoms in each formula unit.
Use your answers to (a), (b), (c) and (d), and the total number of atoms to find the formula of the mineral. [3]

Formula ..............................................................................................

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**END OF PAPER**